



M39N Datasheet

DS-M39N-E01 V1.0

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Contents

Contents	2
1 Features	5
2 Overview	9
3 Pin Descriptions	11
3.1 Pin Descriptions	11
3.2 Pin Multiplexing	13
3.2.1 I/O Pin Predefined States at Reset and After Reset	13
3.2.2 I/O Matrix	14
3.2.3 Pin Multiplexing of Analog Functions	18
4 Peripherals	20
4.1 FEM Support	20
4.2 I/O Multiplexer (IOMUX)	20
4.3 General-purpose I/Os (GPIO)	21
4.4 SPI Interfaces (SPI)	21
4.5 Quad SPI Interface (QSPI)	21
4.6 UART Interfaces (UART)	22
4.7 SDIO Interface (SDIO)	22
4.8 I2C Interfaces (I2C)	23
4.9 GDMA Controller (GDMA)	23
4.10 RGB LED Controller (LED)	24
4.11 PWM Module (PWM)	24
4.12 I2S Interface (I2S)	25
4.13 Auxiliary ADC (AUX ADC)	26
4.14 Timer Groups (TIMG)	26
4.15 Watchdog Timer (WDT)	27
4.16 Real-time Counter (RTC)	27

4.17	Temperature Sensor.....	27
4.18	Anti-tamper Module	27
4.19	Security	27
4.19.1	Secure Boot.....	28
4.19.2	Secure Debug	28
4.19.3	FOTA.....	29
4.19.4	Provisioning	29
4.19.5	Secure Connection	29
4.19.6	BootROM.....	29
4.19.7	Bootloader.....	29
4.19.8	TEE_M.....	30
4.19.9	TrustEngine	30
4.19.9.1	Features	30
4.19.9.2	Supported Standards and Specifications.....	30
4.19.9.3	Components	31
5	Electrical Characteristics.....	33
5.1	Absolute Maximum Ratings	33
5.2	Recommended Operating Conditions	33
5.3	Digital I/O Characteristics.....	33
5.4	Current Consumption.....	34
5.5	RF Characteristics	36
5.5.1	WLAN RF Characteristics.....	36
5.5.1.1	WLAN 2.4 GHz Receiver Characteristics	36
5.5.1.2	WLAN 2.4 GHz Transmitter Characteristics	39
5.5.1.3	WLAN 5 GHz Receiver Characteristics	39
5.5.1.4	WLAN 5 GHz Transmitter Characteristics	43
5.5.2	Bluetooth LE RF Characteristics	44
5.5.2.1	Bluetooth LE Receiver Characteristics	44
5.5.2.2	Bluetooth LE Transmitter Characteristics	46
5.5.3	IEEE 802.15.4 RF Characteristics	48



5.5.3.1	IEEE 802.15.4 Receiver Characteristics	48
5.5.3.2	IEEE 802.15.4 Transmitter Characteristics	49
6	Mechanical Information.....	50
6.1	Module Dimensions.....	50
7	Recommended Land Pattern.....	51
8	Product Handling.....	53
8.1	Moisture Sensitivity Level (MSL).....	53
8.2	Storage Conditions	53
8.3	Baking	53
8.4	Electrostatic Discharge (ESD).....	53
8.5	Reflow Soldering Profile.....	54
9	Ordering Information.....	55
	Revision History	57

1 Features

Module Content

- Dual-band 2.4/5 GHz Wi-Fi 6 (802.11a/b/g/n/ac/ax), Bluetooth Low Energy (LE) 6.0, and IEEE 802.15.4 wireless MCU
- Integrated 40 MHz crystal
- All RF components for transmit and receive matching network
- On-board PCB antenna

Wi-Fi®

- Dual-band 2.4/5 GHz Wi-Fi 6 (IEEE 802.11ax) connectivity, backward compatible with 802.11a/b/g/n/ac standards
- Supports 20 MHz and 40 MHz channel bandwidths on both 2.4 GHz and 5 GHz bands
- 20 MHz channel up to MCS9
- Uplink and downlink Orthogonal Frequency Division Multiple Access (OFDMA)
- Downlink Multi-User Multiple-Input Multiple-Output (DL MU-MIMO)
- Target Wake Time (iTWT/bTWT)
- Dual Carrier Modulation (DCM)
- SU Extended Range (ER)
- STA, SoftAP, Wi-Fi Direct (P2P), and concurrent STA + SoftAP modes
- 802.11mc Fine Timing Measurement (FTM)
- Channel State Information (CSI)
- Supports Wi-Fi, Bluetooth, and 802.15.4 (Zigbee/Thread) coexistence (PTA)
- TX power up to +19 dBm
- RX sensitivity -99 dBm

Bluetooth® Low Energy (LE)

- Bluetooth Core 6.0 compliant
- Support for Bluetooth LE 5.2 features
- Supports Bluetooth LE 1 Mbps, 2 Mbps, and long range (125 kbps and 500 kbps)
- LE Audio with isochronous channels (CIS/BIS)
- Angle of Arrival (AoA) and Angle of Departure (AoD) direction finding, antenna array support up to 16 antennas



- Concurrent roles: Central, Peripheral, Broadcaster, Observer
- Bluetooth Mesh 1.1 support for large-scale IoT networks
- High-power TX mode up to +25 dBm
- RX sensitivity -97 dBm

IEEE 802.15.4

- 2.4 GHz IEEE 802.15.4 compliant
- O-QPSK with 250 kbps data rate
- Thread V1.4.0 support
- Zigbee® 3.0 support
- Matter over Thread

Core

- Arm®v8-M STAR-MC1 core at up to 240 MHz
 - Embedded TrustZone®
 - Supports DSP instructions with SIMD

Memories

- 4 MB flash with XIP support
- 512 KB share SRAM
- 64 KB ROM
- 4 KB OTP (One Time Programmable)

Security

- Secure boot
- Secure debug
- Secure connection
- Firmware Over-The-Air (FOTA)
- Provisioning
- TEE_M
- TrustEngine, including the following features:
 - Symmetric schemes, AES-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM (key size 128-bit, 192-bit and 256-bit)
 - Symmetric schemes, SM4-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM



- Digest schemes, SHA1/224/256
- Digest scheme, SM3
- Asymmetric schemes, RSA 1024/2048/3072/4096 and ECCP 192/224/256/384/512/521
- Asymmetric scheme, SM2
- Key ladder for key management
- Lifecycle management
- True random number generator

Peripherals

- 18 GPIOs
 - Configurable 2.2–3.6 V VIO supply or 1.8–1.95 V VDDRAM supply
- 2x SPI
- 1x QSPI
- 4x UART, 2 with hardware flow control and 1 with flash download support
- 1x SDIO
- 2x I2C
- 1x general-purpose DMA controller (GDMA) with 8 channels
- 1x RGB LED controller
- 12x 32-bit PWM channel
- 1x I2S
- 14-bit AUX ADC, up to 8 channels
- 6x 32-bit general-purpose timer
- 1x watchdog timer
- 1x real-time counter (RTC)
- 1x temperature sensor
- 1x anti-tamper module

Voltage

- Operating voltage/power supply: 2.2 to 5.5 V
- I/O voltage: 2.2–3.6 V VIO supply or 1.8–1.95 V VDDRAM supply

Environmental Specifications

- Temperature
 - Operating temperature: -40 to +85 °C



- Storage temperature: -40 to +125 °C
- Humidity
 - Relative humidity: < 90% noncondensing
 - Storage humidity: < 90% noncondensing

Dimensions

- 33.0 mm x 18.0 mm x 2.5 mm (SMD)

Applications

- Smart home and appliances: door locks, sensors, lighting, and connected appliances
- Consumer electronics: wearables, audio devices with LE Audio, toys, and drones
- Industrial IoT: factory sensors, smart meters, and automation systems
- Healthcare and medical: portable monitors and medical devices
- Edge devices: Matter enabled hubs, controllers, and gateways

2 Overview

The M39N module is built around the highly integrated BK7239N wireless SoC, specifically engineered for next-generation multi-protocol AIoT applications. At its core is a 240 MHz Armv8-M STAR-MC1 processor with built-in TrustZone security technology and DSP extension, along with abundant memory configurations for efficient handling of complex edge computing tasks.

For wireless connectivity, the M39N delivers excellent throughput and low latency through dual-band 2.4/5 GHz Wi-Fi 6 (802.11a/b/g/n/ac/ax), while also supporting the latest Bluetooth LE 6.0 with LE Audio and high-precision positioning capabilities. Furthermore, the integrated IEEE 802.15.4 radio ensures full compatibility with the Zigbee, Thread, and Matter ecosystems, establishing the M39N as a comprehensive cross-protocol connectivity platform.

With its robust feature set and high level of integration, the M39N is the ideal solution for a wide range of connected scenarios, spanning from smart homes, industrial control, to intelligent healthcare.

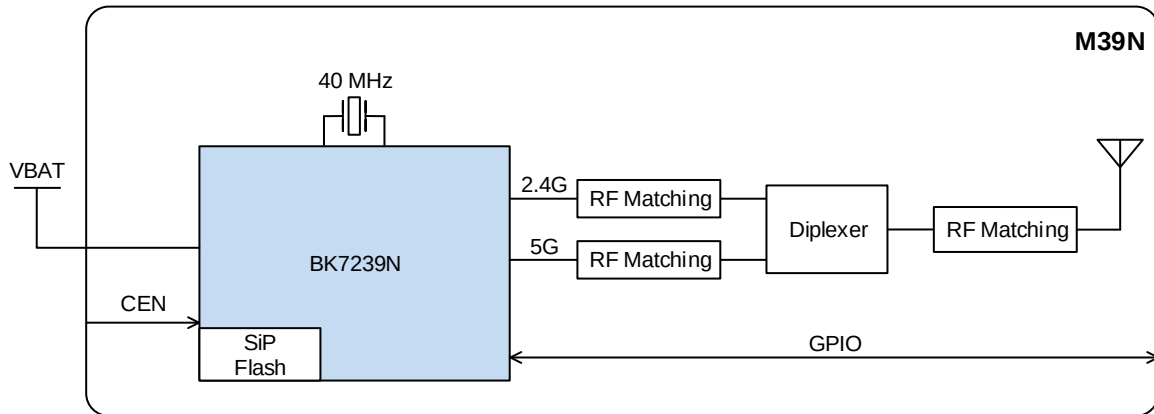
Figure 2-1 illustrates the physical appearance of the M39N module from the top side, featuring the metal shield and PCB antenna.

Figure 2-1 M39N Module Appearance (Top View)



Figure 2-2 shows the general block diagram of the M39N.

Figure 2-2 M39N Block Diagram

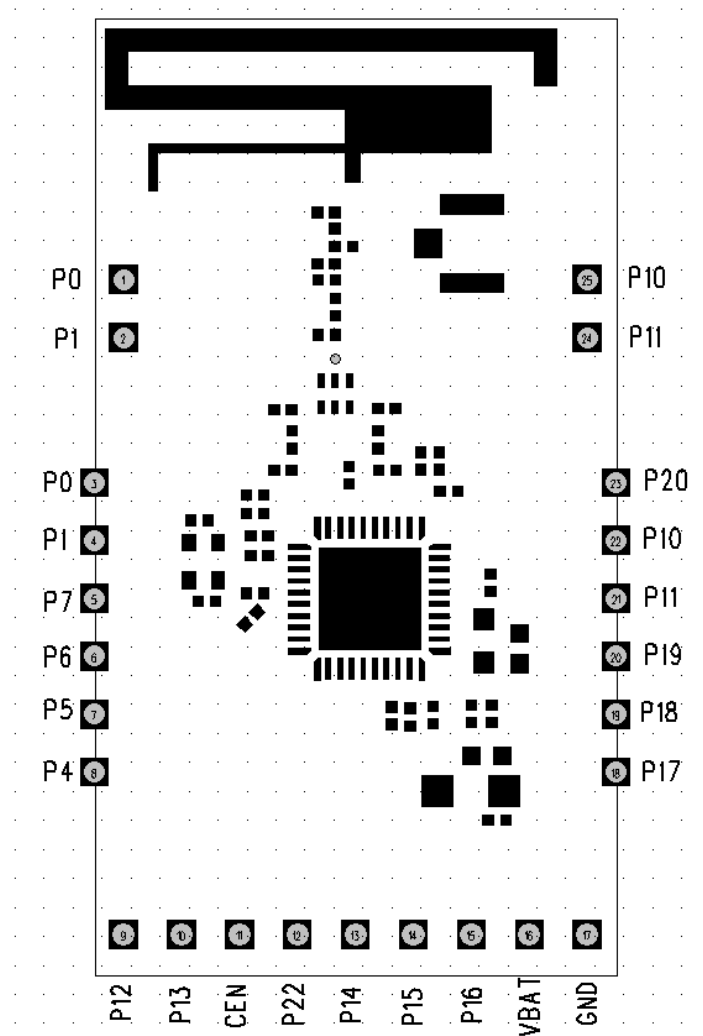


3 Pin Descriptions

3.1 Pin Descriptions

Figure 3-1 illustrates the approximate physical locations and functional assignments of pins on the M39N module.

Figure 3-1 Pin Layout (Top View)



The M39N has 25 pins. Table 3-1 shows the pin descriptions for the M39N.

Note: Due to power-up glitches, the following I/O pins are not recommended for applications sensitive to power-up fluctuations, such as LEDs or motor control: GPIO0, GPIO1, GPIO10, and GPIO11.

Table 3-1 M39N Pin Descriptions

Pin #	Name	I/O	Type	Description
1	P0	I/O	Digital/Analog	- GPIO0: general-purpose I/O - ADC12: analog input channel
2	P1	I/O	Digital/Analog	- GPIO1: general-purpose I/O - ADC13: analog input channel
3	P0	I/O	Digital/Analog	- GPIO0: general-purpose I/O - ADC12: analog input channel
4	P1	I/O	Digital/Analog	- GPIO1: general-purpose I/O - ADC13: analog input channel
5	P7	I/O	Digital/Analog	- GPIO7: general-purpose I/O - ADC6: analog input channel
6	P6	I/O	Digital/Analog	- GPIO6: general-purpose I/O - ADC5: analog input channel
7	P5	I/O	Digital/Analog	- GPIO5: general-purpose I/O - ADC4: analog input channel
8	P4	I/O	Digital/Analog	- GPIO4: general-purpose I/O - ADC3: analog input channel
9	P12	I/O	Digital/Analog	- GPIO12: general-purpose I/O - ADC14: analog input channel
10	P13	I/O	Digital/Analog	- GPIO13: general-purpose I/O - ADC15: analog input channel
11	CEN	-	Analog input	Chip enable, active high
12	P22	I/O	Digital	GPIO22: general-purpose I/O
13	P14	I/O	Digital	GPIO14: general-purpose I/O
14	P15	I/O	Digital	GPIO15: general-purpose I/O
15	P16	I/O	Digital	GPIO16: general-purpose I/O
16	VBAT	-	Power	Module power supply
17	GND	-	GND	Ground
18	P17	I/O	Digital	GPIO17: general-purpose I/O
19	P18	I/O	Digital	GPIO18: general-purpose I/O

Pin #	Name	I/O	Type	Description
20	P19	I/O	Digital	GPIO19: general-purpose I/O
21	P11	I/O	Digital	- GPIO11: general-purpose I/O - DL_UART_TX: UART flash download transmit data output
22	P10	I/O	Digital	- GPIO10: general-purpose I/O - DL_UART_RX: UART flash download receive data input
23	P20	I/O	Digital	GPIO20: general-purpose I/O
24	P11	I/O	Digital	- GPIO11: general-purpose I/O - DL_UART_TX: UART flash download transmit data output
25	P10	I/O	Digital	- GPIO10: general-purpose I/O - DL_UART_RX: UART flash download receive data input

3.2 Pin Multiplexing

Each I/O pin of the M39N can be configured to perform multiple functions: GPIO function, peripheral functions, and analog functions.

The I/O pins can be configured as the following states:

- High impedance
- Input high-impedance
- Input pull-up
- Input pull-down
- Output push-pull with pull-up or pull-down
- Input/output
- Peripheral function push-pull with pull-up or pull-down
- Peripheral function open-drain with pull-up or pull-down
- Analog function

3.2.1 I/O Pin Predefined States at Reset and After Reset

Table 3-2 shows the predefined states at reset and after reset for I/Os.

Table 3-2 I/O Pin Predefined States at Reset and After Reset

I/O	Predefined State	
	At Reset	After Reset
GPIO0	Peripheral function (pull-down) (SWCLK)	Peripheral function (pull-down) (SWCLK)
GPIO1	Peripheral function (pull-up) (SWDIO)	Peripheral function (pull-up) (SWDIO)
GPIO4	High impedance	High impedance
GPIO5	High impedance	High impedance
GPIO6	High impedance	High impedance
GPIO7	High impedance	High impedance
GPIO10	Input pull-down	Peripheral function (DL_UART_RX)
GPIO11	Input pull-down	High impedance
GPIO12	High impedance	High impedance
GPIO13	High impedance	High impedance
GPIO14	High impedance	High impedance
GPIO15	High impedance	High impedance
GPIO16	High impedance	High impedance
GPIO17	High impedance	High impedance
GPIO18	High impedance	High impedance
GPIO19	High impedance	High impedance
GPIO20	High impedance	High impedance
GPIO22	High impedance	High impedance

3.2.2 I/O Matrix

The I/O multiplexer provides one configuration register for each I/O pin. Each I/O pin can be configured as a GPIO or an I/O for peripherals through the GPIOx_FUNC_SEL[6:0] field of the configuration register. Table 3-3 shows the I/O matrix.

Table 3-3 I/O Matrix

Function	GPIOx_FUNC_SEL [6:0] Value	I/O Function Name	I/O Type	Description
GPIO	7'd0	GPIO	High impedance	General-purpose I/O
	7'd1	GPIO	Input	General-purpose I/O
	7'd2	GPIO	Output	General-purpose I/O
	7'd3	GPIO	Input/output	General-purpose I/O
PTA	7'd4	BT_ACTIVE	Input	Bluetooth active
	7'd9	BT_PRIORITY	Input	Bluetooth priority
	7'd92	WIFI_ACTIVE	Output	Wi-Fi active
AoA/AoD	7'd5	BT_ANT0	Output	Bluetooth antenna select
	7'd6	BT_ANT1	Output	Bluetooth antenna select
	7'd7	BT_ANT2	Output	Bluetooth antenna select
	7'd8	BT_ANT3	Output	Bluetooth antenna select
Clock	7'd10	CLK_AUXS	Output	Clock output derived from 40 MHz CLK_XTAL/32 kHz CLK_32K/480 MHz CLK_DPLL480M (divide by 1 to 16)
	7'd11	CLK_XTALO	Output	40 MHz clock output
	7'd12	CLK_XTAL_DIV	Output	40 MHz clock output (divide by 1/2/4/8)
	7'd40	LPO_CLK	Output	32 kHz clock output
FEM support	7'd29	FEM_LNA_EN	Output	LNA bypass
	7'd93	WIFI_RX_EN	Output	Receive enable
	7'd94	WIFI_TX_EN	Output	Transmit enable
I2C0	7'd30	I2C0_SCL	Input/output	Serial clock
	7'd31	I2C0_SDA	Input/output	Serial data

Function	GPIOx_FUNC_SEL [6:0] Value	I/O Function Name	I/O Type	Description
I2C1	7'd32	I2C1_SCL	Input/output	Serial clock
	7'd33	I2C1_SDA	Input/output	Serial data
I2S	7'd34	I2S_MCLK	Output	Master clock
	7'd35	I2S_DIN	Input	Serial data input
	7'd36	I2S_DOUT	Output	Serial data output
	7'd37	I2S_SCK	Input/output	Serial clock
	7'd38	I2S_SYNC	Input/output	Frame synchronization
LED	7'd39	LED	Output	LED output
PWM	7'd44	PWM0	Input/output	PWM0 channel
	7'd45	PWM1	Input/output	PWM1 channel
	7'd46	PWM2	Input/output	PWM2 channel
	7'd47	PWM3	Input/output	PWM3 channel
	7'd48	PWM4	Input/output	PWM4 channel
	7'd49	PWM5	Input/output	PWM5 channel
	7'd50	PWM6	Input/output	PWM6 channel
	7'd51	PWM7	Input/output	PWM7 channel
	7'd52	PWM8	Input/output	PWM8 channel
	7'd53	PWM9	Input/output	PWM9 channel
	7'd54	PWM10	Input/output	PWM10 channel
	7'd55	PWM11	Input/output	PWM11 channel
QSPI	7'd56	QSPI_CS	Output	Chip select
	7'd57	QSPI_IO0	Input/output	Data
	7'd58	QSPI_IO1	Input/output	Data
	7'd59	QSPI_IO2	Input/output	Data
	7'd60	QSPI_IO3	Input/output	Data
	7'd61	QSPI_SCK	Input/output	Serial clock
SDIO	7'd62	SDIO_CLK	Input/output	Clock

Function	GPIOx_FUNC_SEL [6:0] Value	I/O Function Name	I/O Type	Description
	7'd63	SDIO_CMD	Input/output	Command/response
	7'd64	SDIO_DATA0	Input/output	Data
	7'd65	SDIO_DATA1	Input/output	Data
	7'd66	SDIO_DATA2	Input/output	Data
	7'd67	SDIO_DATA3	Input/output	Data
SPI0	7'd68	SPI0_MISO	Input/output	Master in slave out
	7'd69	SPI0_MOSI	Input/output	Master out slave in
	7'd70	SPI0_CSN	Input/output	Chip select
	7'd71	SPI0_SCK	Input/output	Serial clock
SPI1	7'd72	SPI1_MISO	Input/output	Master in slave out
	7'd73	SPI1_MOSI	Input/output	Master out slave in
	7'd74	SPI1_CSN	Input/output	Chip select
	7'd75	SPI1_SCK	Input/output	Serial clock
SWD	7'd76	SWCLK	Input	Serial wire clock
	7'd77	SWDIO	Input/output	Serial wire data
Anti-tamper	7'd78	TAMP_RX	Input	Receive data input
	7'd79	TAMP_TX	Output	Transmit data output
UART0	7'd80	UART0_CTS	Input	Clear to send
	7'd81	UART0_RTS	Output	Request to send
	7'd82	UART0_RX	Input	Receive data input
	7'd83	UART0_TX	Output	Transmit data output
UART1	7'd84	UART1_RX	Input	Receive data input
	7'd85	UART1_TX	Output	Transmit data output
UART2	7'd86	UART2_RX	Input	Receive data input
	7'd87	UART2_TX	Output	Transmit data output
UART3	7'd88	UART3_CTS	Input	Clear to send
	7'd89	UART3_RTS	Output	Request to send

Function	GPIOx_FUNC_SEL [6:0] Value	I/O Function Name	I/O Type	Description
	7'd90	UART3_RX	Input	Receive data input
	7'd91	UART3_TX	Output	Transmit data output

3.2.3 Pin Multiplexing of Analog Functions

Table 3-4 shows the pin multiplexing of analog functions. The I/O pins must be set to input pull-up or input pull-down to enable analog functions.

All GPIOs can be configured as wake-up (WKUP) function to wake the system from deep sleep and sleep modes.

Table 3-4 Pin Multiplexing of Analog Functions

GPIO	Analog Function
	AUX ADC
GPIO0	ADC12
GPIO1	ADC13
GPIO4	ADC3
GPIO5	ADC4
GPIO6	ADC5
GPIO7	ADC6
GPIO10	
GPIO11	
GPIO12	ADC14
GPIO13	ADC15
GPIO14	
GPIO15	
GPIO16	
GPIO17	
GPIO18	
GPIO19	
GPIO20	



GPIO	Analog Function
	AUX ADC
GPIO22	

4 Peripherals

4.1 FEM Support

The M39N supports the use of an external Front-End Module (FEM). The external FEM is controlled by three signals WIFI_TX_EN, WIFI_RX_EN, and FEM_LNA_EN.

Table 4-1 FEM Operating Modes

Operating Mode	WIFI_TX_EN	WIFI_RX_EN	FEM_LNA_EN
Transmit mode	1	0	0
Receive mode	0	1	1
LNA bypass mode	0	1	0
	1	1	X
Standby mode	0	0	0
Not supported	All other states		

4.2 I/O Multiplexer (IOMUX)

The M39N has an I/O multiplexer (IOMUX) that allows the use of I/O pins as either GPIOs or I/Os for peripherals. The I/O multiplexer provides one configuration register for each I/O pin. The GPIOx_FUNC_SEL[6:0] field of the configuration register selects I/O functions:

- When GPIOx_FUNC_SEL[6:0] is set to 0 to 3, the I/O is configured as a GPIO. The GPIO type can be configured as:
 - High impedance
 - Input
 - Output
 - Input/output
- When GPIOx_FUNC_SEL[6:0] is set to 4 to 94, the I/O is configured as an I/O of a peripheral. It's should be noted that multiple I/Os cannot be configured as the same peripheral I/O.

The I/O multiplexer provides additional registers that, in GPIO mode, allow for simultaneous querying of the interrupt status and input status of all GPIOs, as well as simultaneous configuration of the output of all GPIOs.

4.3 General-purpose I/Os (GPIO)

The M39N has up to 18 GPIOs, which can be configured as high-impedance, input, output, or input/output. All GPIOs are shared with peripheral and/or analog functions.

The main features of GPIOs include:

- Push-pull
- Internal pull-up/down resistors
- Configurable drive strength
- Peripheral function
- Analog function
- Interrupt generation:
 - High or low level
 - Rising or falling edge
- Configurable 2.2–3.6 V VIO supply or 1.8–1.95 V VDDRAM supply

4.4 SPI Interfaces (SPI)

The M39N integrates two SPI interfaces that can operate in master or slave mode. The SPI interfaces allow a clock frequency up to 40 MHz in both master and slave modes.

The SPI interfaces support the following features:

- 4-wire or 3-wire full-duplex synchronous communication
- Configurable 8-bit or 16-bit data width
- Programmable clock polarity and phase
- Programmable data order with MSB-first or LSB-first shifting
- Embedded 64-depth RX FIFO and 64-depth TX FIFO with DMA capability

4.5 Quad SPI Interface (QSPI)

The M39N embeds a Quad SPI interface that provides support for communicating with external flash, PSRAM, or AMOLED displays. The QSPI interface allows communicating up to 80 MHz.

The features of the QSPI interface are listed below:

- Single, dual, or quad SPI input/output

- Two functional modes: indirect mode and memory-mapped mode
- Fully programmable opcode and frame format
- Integrated RX FIFO and TX FIFO
- Supports 8, 16, and 32-bit data accesses

4.6 UART Interfaces (UART)

The M39N includes four Universal Asynchronous Receiver/Transmitter (UART) interfaces, which support full-duplex, asynchronous serial communication at a baud rate up to 10 Mbps.

UART0 supports flash download. When UART0 is used for flash download, UART0_RX and UART0_TX are mapped to GPIO10 as DL_UART_RX and GPIO11 as DL_UART_TX, respectively.

The UART interfaces offer the features below:

- Configurable data length (5, 6, 7, or 8 bits)
- Even, odd, or none parity check
- Programmable stop bits (1 or 2 bits)
- Each UART embeds a 128-byte TX FIFO and a 128-byte RX FIFO. FIFO mode is disabled by default and can be enabled by software.
- Hardware flow control with RTS and CTS signals (UART0 and UART3)
- Flash download (UART0)
- Programmable digital filter

4.7 SDIO Interface (SDIO)

A secure digital input/output (SDIO) host/slave interface is available on the M39N. It can be used as a host to read external SD cards or used by an external host as a slave to communicate with chips. The SDIO interface allows a maximum clock speed of 80 MHz.

The SDIO features include the following:

- SD memory card specification version 2.0 compliant
- SDIO card specification version 2.0 compliant
- Two data bus modes: 1-bit mode (default) and 4-bit mode
- Data transfer up to 40 Mbyte/s for the host mode and 20 Mbyte/s for the slave mode
- Supports DMA capability, allowing high-speed transfer without CPU load

4.8 I2C Interfaces (I2C)

I2C is a popular inter-IC interface that requires only two bus lines, the serial data line (SDA) and the serial clock line (SCL). The M39N embeds two I2C interfaces, which can operate in master or slave mode.

The features of the I2C interfaces are listed below:

- Master and slave modes
- Standard-mode (up to 100 kbps)
- Fast-mode (up to 400 kbps)
- Fast-mode Plus (up to 1 Mbps)
- 7-bit and 10-bit addressing
- Bus idle and SCL low timeout condition detection
- Embedded 16-byte TX FIFO and 16-byte RX FIFO

4.9 GDMA Controller (GDMA)

The M39N has a general-purpose DMA controller (GDMA) with 8 DMA channels to unload CPU activity. The 8 channels are shared by peripherals that have DMA capabilities.

The GDMA controller has the features below:

- Supports transfer of the following types:
 - Single block transfer
 - Repeated block transfer
- Memory-mapped data transfers from a source to a destination:
 - Peripheral-to-memory
 - Memory-to-peripheral
 - Memory-to-memory
 - Peripheral-to-peripheral
- Programmable data width for destination and source:
 - 8 bits (byte)
 - 16 bits (half-word)
 - 32 bits (word)
- Eight-grade priority for transfers arbitration at a channel level
- Per channel event generation on the following events:

- Transfer complete
 - Half transfer complete
 - Bus error
- Each channel can be independently configured to generate interrupts for specific events.
- 8 concurrent DMA channels:
 - Each channel has a FIFO for queuing source and destination transfers.
- Secure/non-secure support:
 - A non-secure channel performs non-secure DMA transfers.
 - A secure channel can perform secure or non-secure DMA transfers.
 - TrustZone-aware DMA AHB master port
- Privileged/unprivileged support:
 - An unprivileged channel performs unprivileged DMA transfers.
 - A privileged channel performs privileged DMA transfers.
 - Privileged-aware AHB master port
 - Privileged-aware AHB slave port

A selection of peripherals on the M39N have DMA capabilities, including UART0, UART1, UART2, UART3, SPI0, SPI1, SDIO, and I2S.

4.10 RGB LED Controller (LED)

The M39N has a RGB LED controller that supports Return-to-Zero (RZ) communication protocol. With an external LED driver IC, the RGB LED controller can support up to 1024 LED beads. The duty cycle of code 0 and code 1 of RZ communication can be configured by register. The reset signal length is also configurable.

4.11 PWM Module (PWM)

The M39N has an advanced-control PWM module (PWM). The PWM module consists of 12 32-bit auto-reload counters driven by 12 programmable prescalers. The PWM module has 12 independent channels for input capture, pulse edge counting, or generation of output waveforms (output compare).

The 12 independent channels can be allocated to any counter comparator output. Each of the 12 channels can generate independent waveforms or complementary waveforms when coupled with any other channels.

The features of the PWM module are listed here:

- 12 32-bit up, down, up-and-down auto-reload counters

- 12 8-bit programmable prescalers capable of dividing the clock frequency of each counter by any factor between 1 and 256
- 12 independent channels for:
 - Input capture
 - Pulse edge counting
 - PWM generation (edge or center-aligned mode)
- Complementary outputs with programmable dead-time and configurable dead-time mode
- Synchronization circuit to control the counter with external signals and to interconnect several counters together
- Repetition counter to update the registers only after a given number of cycles of the counter
- Dedicated hardware support for smooth and precise LED dimming/fading
- Interrupt generation on the following events:
 - Update: counter overflow or underflow, counter initialization (by software or internal/external trigger)
 - Counter start
 - Input capture
 - Output compare
 - LED fading end
- Change of polarity, duty cycle, and base frequency on every PWM period
- Supports incremental (quadrature) encoder and Hall-sensor circuitry for positioning purposes

4.12 I2S Interface (I2S)

The M39N integrates an I2S interface that supports master and slave modes with a sampling rate from 8 kHz to 384 kHz. The I2S interface supports both PCM mono channel mode and I2S stereo channel mode.

Listed here are the I2S features:

- Master or slave mode
- Full duplex or half-duplex communication
- Various sampling rates
- 12-bit programmable prescaler
- Multiple I2S protocols supported:
 - I2S Philips standard
 - MSB-Justified standard (Left-Justified)
 - LSB-Justified standard (Right-Justified)

- PCM standard
- Programmable data order with LSB first or MSB first
- Programmable data width between 1 and 32 bits
- Programmable clock polarity
- Integrated 32-bit RX FIFO and 32-bit TX FIFO, each with a depth of 32 words
- Master clock can be output to drive external audio devices.

4.13 Auxiliary ADC (AUX ADC)

The auxiliary ADC (AUX ADC) is a 14-bit successive approximation analog-to-digital converter. The AUX ADC has multiple external analog input channels and internal dedicated channels. The AUX ADC supports A/D conversion performed in one-shot or continuous mode.

The AUX ADC module has the following features:

- Programmable sampling rates from 51.28 kHz to 1.538 MHz
- 14-bit resolution
- Up to 8 external analog input channels: ADC3/4/5/6/12/13/14/15
- Five internal dedicated channels:
 - VBAT monitoring channel (VBAT/4), connected to ADC0
 - Internal temperature sensor (VTEMP), connected to ADC7
 - Two internal calibration channels, connected to ADC8/11
 - Internal debug channel, connected to ADC9
- Conversion modes:
 - One-shot mode
 - Continuous mode

4.14 Timer Groups (TIMG)

The M39N includes two general-purpose timer groups (TIMG). Each group has three 32-bit timers. Each group consists of three 32-bit counters driven by a 4-bit prescaler.

Each TIMG module has the following features:

- Three timers (Timer0/1/2)
- Three 32-bit up counters
- 4-bit prescaler, division factor between 1 and 16

- The real-time value of the counter can be read.

4.15 Watchdog Timer (WDT)

The purpose of the watchdog timer is to detect and recover from failures or malfunctions. The watchdog timer triggers a reset on expiry of a specified time period.

The watchdog timer runs on a 16 kHz clock derived from CLK_32K (CLK_32K/2). The division ratio for the 16 kHz clock can be 2, 4, 8, or 16, allowing for a maximum programmable period of up to 65.536 ($2^{16}/1$ kHz) seconds.

4.16 Real-time Counter (RTC)

The real-time counter (RTC) module features a 36-bit counter and a tick event generator. The RTC clock source can be selected from CLK_X32K or CLK_ROSC. It is used for low-power timing, and it keeps running even when the system is in deep sleep0 mode.

4.17 Temperature Sensor

The M39N integrates an on-chip temperature sensor that can measure on-chip temperature over -40 to +125 °C with an accuracy of ± 5 °C. The digital results can be read from the AUX ADC.

Usually, the software initiates the calibration of a specific module based on the temperature value, narrowing the difference in chip performance at different temperatures. The host can also read the on-chip temperature and decide whether to reduce transmit power or suspend operation at high temperatures.

4.18 Anti-tamper Module

The anti-tamper module is designed to prevent, detect, and respond to physical attacks on the M39N. Once a tampering attempt is detected, it reports an interrupt to the MCU.

4.19 Security

The M39N provides state-of-the-art security based on a powerful security architecture. It integrates a Secure Element (SE) for cryptograph and system security control. The SE embeds all-round and robust security features to set up a top-secret execution environment for IoT devices.

The key features of the SE include:

- Secure boot
- Secure debug
- Secure connection
- Firmware Over-The-Air (FOTA)
- Provisioning
- TEE_M
- TrustEngine

System Architecture

The SE software implements an all-round device-to-cloud security architecture, including:

- Provisioning mechanism in the manufacturing stage.
- Secure boot, secure debug, TRNG calibration, and FOTA update in the device boot stage.
- Secure connection and FOTA download in the device run-time stage.

The SE also introduces a remote management solution to enable users to access, monitor and manage IoT devices in a secure manner. The remote management solution includes FOTA SaaS, Secure Debug SaaS, and Provisioning SaaS.

4.19.1 Secure Boot

The secure boot solution provides the legitimacy and trustworthiness for images running on devices. It ensures that only official images which are published from OEMs are executable.

The secure boot functions include following aspects:

- Image verification
- Image encryption
- Extended program execution
- Image anti-rollback protection

4.19.2 Secure Debug

Secure debug provides a reliable mechanism for enabling the debug feature of a device after the device deployment.

Debug is one of the most commonly-used features in the device. With the debug feature, you can access all the device data, including the device firmware and the device root key.

4.19.3 FOTA

The FOTA solution implements a series of light-weight and trustworthy firmware upgrade interaction between IoT devices and servers, which includes downloading, verification, and installation.

4.19.4 Provisioning

Provisioning is the process to initialize secure credentials onto devices. It is required to happen in a secure environment such as the product line in the manufacturing stage.

The provisioning solution provides a full secure mechanism from the cloud server to devices. It keeps the security and integrity of provisioning materials, such as the device ID, model key, and the secure boot/secure debug public key hash, also keeps the device top secret such as the device root key that occurs only on the device side.

4.19.5 Secure Connection

The secure connection feature uses Mbed TLS to provide connection security in the transport layer over UDP. The key exchange method is Pre-Shared Key (PSK). The PSK's deployment happens in the manufacturing stage.

Mbed TLS provides cryptograph and SSL/TLS capabilities with TLS 1.1 and TLS 1.2.

4.19.6 BootROM

The SE includes the BootROM reference code which is integrated with the secure boot solution.

There are primary boot and recovery boot paths in the BootROM.

- Primary boot is the main boot path. Normally devices should boot to the primary path.
- Recovery boot is the secondary boot path.

4.19.7 Bootloader

The SE includes the Bootloader reference code which is integrated with the secure debug, provisioning, TRNG calibration, and FOTA solution.

In the primary bootloader, the secure debug agent, provisioning agent, and TRNG agent are integrated. The primary bootloader boots to the next image—TEE_M.

In the recovery bootloader, both the secure debug agent, TRNG agent, and the FOTA update agent are integrated. The recovery bootloader does not boot to any images.

4.19.8 TEE_M

The SE introduces TEE_M. TEE_M provides an implementation of secure world software for Armv8-M, which follows Arm Platform Security Architecture (PSA) PSA_Firmware_Framework_1.0-bet0. TEE_M provides the following services by default:

- Secure Storage: The Secure Storage (Only supports Protected Storage) service supports PSA_Storage_API-1.0-bet2 without extended functions.
- Crypto: The Crypto service supports PSA_Cryptography_API_Reference_1.0_bet1.
- DTLS: DTLS (leverages Mbed TLS) is integrated in TEE_M.

4.19.9 TrustEngine

The SE introduces TrustEngine as a secure component in the system. It provides the following features:

- High security assurance. The crypto engine supports key ladders, lifecycle management and True Random Number Generator (TRNG) which enhance the system security.
- High performance and low power for encryption/decryption operation. This is achieved by TrustEngine internal cryptography engines.
- Reduction of software complicity in security. Some security functions are implemented in TrustEngine hardware, which can reduce the risk of sensitive information leakage to non-secure hosts.

4.19.9.1 Features

TrustEngine includes the following features:

- Symmetric schemes, AES-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM (key size 128-bit, 192-bit and 256-bit)
- Symmetric schemes, SM4-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM
- Digest schemes, SHA1/224/256
- Digest scheme, SM3
- Asymmetric schemes, RSA 1024/2048/3072/4096 and ECCP 192/224/256/384/512/521
- Asymmetric scheme, SM2
- Key ladder for key management
- Lifecycle management
- True random number generator

4.19.9.2 Supported Standards and Specifications

TrustEngine is compliant with the following standards:

- FIPS PUB 180-4: Secure Hash Standard (SHS)
- FIPS PUB 197: Advanced Encryption Standard (AES)
- NIST SP 800-38A Recommendation for Block Cipher Modes of Operation-Methods and Techniques
- NIST SP 800-38B Recommendation for Block Cipher Modes of Operation: The CMAC Mode for Authentication
- NIST SP 800-38C Recommendation for Block Cipher Modes of Operation-the CCM Mode for Authentication and Confidentiality
- NIST SP 800-38D, Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC
- NIST SP 800-90B Recommendation for the Entropy Sources Used for Random Bit Generation
- SM2 Public Key Cryptographic Algorithm Based on Elliptic Curves (GB/T 32918-2016)
- SM3 Cryptographic Hash Algorithm (GB/T 32905-2016)
- SM4 Block Cipher Algorithm (GB/T 32907-2016)

4.19.9.3 Components

TrustEngine consists of five major function blocks.

- Symmetric Cryptography Accelerator (SCA)
- Asymmetric Cryptography Accelerator (ACA)
- HASH engine
- One-time programmable storage access controller
- True random number generator

Symmetric Cryptography Accelerator (SCA)

SCA in TrustEngine is responsible for data encryption/decryption using the symmetric cryptography algorithm. The data encryption/decryption operations are performed through the SCA engine by sending special commands to TrustEngine.

Asymmetric Cryptography Accelerator (ACA)

The ACA engine in TrustEngine is responsible for accelerating the asymmetric cryptography, such as:

- Asymmetric encryption and decryption: RSA and ECC
- Digital signature and verification: RSA signatures and ECDSA
- Key exchange: DH (Diffie-Hellman) and ECDH

HASH Accelerator

The HASH engine in TrustEngine is responsible for digest calculation. The digest of certain data can be calculated through the HASH engine by sending special commands to TrustEngine.

True Random Number Generator

The True Random Number Generator (TRNG) generates the random bits using the internal entropy source (ring oscillator inverter chain) or the external entropy source. The random bits are required by both secure and non-secure hosts.

5 Electrical Characteristics

5.1 Absolute Maximum Ratings

Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. Exposure to absolute-maximum-rated conditions for prolonged periods may affect the reliability of the device.

Parameter	Description	Min.	Max.	Unit
V _{BAT}	Power supply voltage	-0.3	6.0	V
V _{IO}	I/O voltage	-0.3	4.0	V
V _{DDRAM}	I/O voltage	-0.3	2.1	V
T _{STR}	Storage temperature range	-40	125	°C

5.2 Recommended Operating Conditions

Parameter	Description	Min. ⁽¹⁾	Typ.	Max.	Unit
V _{BAT} ⁽²⁾	Power supply voltage	2.2	3.3	5.5	V
V _{IO}	I/O voltage	2.2	3.0	3.6	V
V _{DDRAM}	I/O voltage	1.8	-	1.95	V
T _{OPR}	Operating temperature range	-40	-	85	°C

(1) The minimum voltage specified includes the ripple on the supply voltage and all other transient dips. Care must be taken when operating at the minimum specified voltage.

(2) To ensure WLAN performance, the ripple on the supply must be less than V_{pp} = 100 mV.

5.3 Digital I/O Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{IH}	High-level input voltage	-	0.7 V _{IO}	-	V _{IO} + 0.3	V
V _{IL}	Low-level input voltage	-	-0.3	-	0.3 V _{IO}	V
V _{OH}	High-level output voltage	-	0.8 V _{IO}	-	-	V
V _{OL}	Low-level output voltage	-	-	-	0.1 V _{IO}	V
I _{OH}	High-level source current	V _{OH} = 0.8 V _{IO} , GPIO_CAPACITY = 3	-	40	-	mA

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
		VOH = 0.8 VIO, GPIO_CAPACITY = 2	-	30	-	mA
		VOH = 0.8 VIO, GPIO_CAPACITY = 1	-	18	-	mA
		VOH = 0.8 VIO, GPIO_CAPACITY = 0	-	10	-	mA
I _{OL}	Low-level sink current	VOL = 0.15 VIO, GPIO_CAPACITY = 3	-	36	-	mA
		VOL = 0.15 VIO, GPIO_CAPACITY = 2	-	28	-	mA
		VOL = 0.15 VIO, GPIO_CAPACITY = 1	-	16	-	mA
		VOL = 0.15 VIO, GPIO_CAPACITY = 0	-	8	-	mA
R _{PU}	Weak pull-up resistor	-	-	40	-	kΩ
R _{PD}	Weak pull-down resistor	-	-	44	-	kΩ

5.4 Current Consumption

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Band	Condition	Min.	Typ.	Max.	Unit
Active Mode						
RX current	2.4 GHz	-	-	12	-	mA
	5 GHz	-	-	16	-	mA
TX current	WLAN 2.4 GHz	11b: 11 Mbps DSSS @ 17 dBm	-	270	-	mA
		11g: 54 Mbps OFDM @ 15 dBm	-	255	-	mA
		11n: MCS7, HT20 @ 14 dBm	-	250	-	mA
		11ax: MCS7, HE20 @ 14 dBm	-	235	-	mA
	WLAN 5 GHz	11a: 54 Mbps OFDM @ 15 dBm	-	300	-	mA
		11n: MCS7, HT20 @ 14 dBm	-	295	-	mA
		11ax: MCS7, HE20 @ 14 dBm	-	285	-	mA

Parameter	Band	Condition	Min.	Typ.	Max.	Unit
	Bluetooth LE	15 dBm	-	210	-	mA
		6 dBm	-	120	-	mA
		0 dBm	-	100	-	mA
		-15 dBm	-	90	-	mA
Wi-Fi IEEE Power Save Mode						
Standby associated, DTIM = 1	2.4 GHz	Without TCP keep-alive	-	675	-	μA
Standby associated, DTIM = 3		Without TCP keep-alive	-	255	-	μA
Standby associated, DTIM = 10		Without TCP keep-alive	-	102	-	μA
Standby associated, DTIM = 1	5 GHz	Without TCP keep-alive	-	655	-	μA
Standby associated, DTIM = 3		Without TCP keep-alive	-	240	-	μA
Standby associated, DTIM = 10		Without TCP keep-alive	-	100	-	μA
Sleep Mode						
Deep sleep0		Only RTC is active	-	2.5	-	μA
Deep sleep1		128 KB SRAM retention	-	13	-	μA
Sleep		-	-	47	-	μA
Shutdown Mode						
Shutdown		-	-	0.5	-	μA

5.5 RF Characteristics

Unless otherwise noted, the RF data is measured at the antenna port, where RF cable is connected, including the front-end loss. The front-end circuit is a matching network designed to ensure that the product complies with wireless regulations in various countries.

5.5.1 WLAN RF Characteristics

5.5.1.1 WLAN 2.4 GHz Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2412	-	2484	MHz
Sensitivity					
Sensitivity - IEEE 802.11b (8% PER)	1 Mbps DSSS	-	-99	-	dBm
	2 Mbps DSSS	-	-96	-	dBm
	5.5 Mbps DSSS	-	-94	-	dBm
	11 Mbps DSSS	-	-90	-	dBm
Sensitivity - IEEE 802.11g (10% PER)	6 Mbps OFDM	-	-93	-	dBm
	9 Mbps OFDM	-	-92	-	dBm
	12 Mbps OFDM	-	-91	-	dBm
	18 Mbps OFDM	-	-89	-	dBm
	24 Mbps OFDM	-	-86	-	dBm
	36 Mbps OFDM	-	-83	-	dBm
	48 Mbps OFDM	-	-79	-	dBm
	54 Mbps OFDM	-	-77	-	dBm
Sensitivity - IEEE 802.11n, 20 MHz (10% PER)	HT20, MCS0	-	-93	-	dBm
	HT20, MCS1	-	-91	-	dBm
	HT20, MCS2	-	-88	-	dBm
	HT20, MCS3	-	-86	-	dBm

Parameter	Condition	Min.	Typ.	Max.	Unit
	HT20, MCS4	-	-82	-	dBm
	HT20, MCS5	-	-78	-	dBm
	HT20, MCS6	-	-77	-	dBm
	HT20, MCS7	-	-75	-	dBm
Sensitivity - IEEE 802.11n, 40 MHz (10% PER)	HT40, MCS0	-	-89	-	dBm
	HT40, MCS1	-	-87	-	dBm
	HT40, MCS2	-	-85	-	dBm
	HT40, MCS3	-	-83	-	dBm
	HT40, MCS4	-	-79	-	dBm
	HT40, MCS5	-	-76	-	dBm
	HT40, MCS6	-	-74	-	dBm
	HT40, MCS7	-	-72	-	dBm
Sensitivity - IEEE 802.11ax, 20 MHz (10% PER)	HE20, MCS0	-	-93	-	dBm
	HE20, MCS1	-	-91	-	dBm
	HE20, MCS2	-	-89	-	dBm
	HE20, MCS3	-	-86	-	dBm
	HE20, MCS4	-	-82	-	dBm
	HE20, MCS5	-	-79	-	dBm
	HE20, MCS6	-	-77	-	dBm
	HE20, MCS7	-	-75	-	dBm
	HE20, MCS8	-	-72	-	dBm
	HE20, MCS9	-	-69	-	dBm
Maximum Receive Level					
Maximum receive level @ 2.4 GHz	11b: 1 Mbps (8% PER, 1024 octets)	-	10	-	dBm
	11b: 11 Mbps (8% PER, 1024 octets)	-	10	-	dBm
	11g: 6 Mbps (10% PER, 1000 octets)	-	10	-	dBm

Parameter	Condition		Min.	Typ.	Max.	Unit
	11g: 54 Mbps (10% PER, 1000 octets)		-	5	-	dBm
	11n: HT20 MCS0 (10% PER, 4096 octets)		-	10	-	dBm
	11n: HT20 MCS7 (10% PER, 4096 octets)		-	0	-	dBm
	11n: HT40 MCS0 (10% PER, 4096 octets)		-	10	-	dBm
	11n: HT40 MCS7 (10% PER, 4096 octets)		-	0	-	dBm
	11ax: HE20 MCS0 (10% PER, 4096 octets)		-	10	-	dBm
	11ax: HE20 MCS9 (10% PER, 4096 octets)		-	0	-	dBm
Adjacent Channel Rejection						
Adjacent channel (±30 MHz) rejection - IEEE 802.11b (8% PER with desired signal level as specified in Condition)	1 Mbps DSSS	-74 dBm	35	44	-	dB
	2 Mbps DSSS	-74 dBm	35	44	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11b (8% PER with desired signal level as specified in Condition)	5.5 Mbps DSSS	-70 dBm	35	41	-	dB
	11 Mbps DSSS	-70 dBm	35	40	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11g (10% PER with desired signal level as specified in Condition)	6 Mbps OFDM	-79 dBm	16	43	-	dB
	54 Mbps OFDM	-62 dBm	-1	24	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Condition)	HT20, MCS0	-79 dBm	16	40	-	dB
	HT20, MCS7	-61 dBm	-2	20	-	dB
Adjacent channel (±40 MHz) rejection - IEEE 802.11n	HT40, MCS0	-76 dBm	16	33	-	dB

Parameter	Condition		Min.	Typ.	Max.	Unit
(10% PER with desired signal level as specified in Condition)	HT40, MCS7	-58 dBm	-2	19	-	dB
Adjacent channel (± 20 MHz) rejection - IEEE 802.11ax (10% PER with desired signal level as specified in Condition)	HE20, MCS0	-79 dBm	16	40	-	dB
	HE20, MCS9	-54 dBm	-9	14	-	dB

5.5.1.2 WLAN 2.4 GHz Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2412	-	2484	MHz
TX Power					
TX power - IEEE 802.11b (SEM compliant)	1 Mbps DSSS	-	19	-	dBm
	11 Mbps DSSS	-	19	-	dBm
TX power - IEEE 802.11g (EVM compliant)	6 Mbps OFDM	-	18	-	dBm
	54 Mbps OFDM	-	16	-	dBm
TX power - IEEE 802.11n (EVM compliant)	HT20, MCS0	-	17	-	dBm
	HT20, MCS7	-	16	-	dBm
	HT40, MCS0	-	15	-	dBm
	HT40, MCS7	-	13	-	dBm
TX power - IEEE 802.11ax (EVM compliant)	HE20, MCS0	-	17	-	dBm
	HE20, MCS9	-	13	-	dBm

5.5.1.3 WLAN 5 GHz Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	4900	-	5925	MHz

Parameter	Condition	Min.	Typ.	Max.	Unit
Sensitivity					
Sensitivity - IEEE 802.11a (10% PER)	6 Mbps OFDM	-	-91	-	dBm
	9 Mbps OFDM	-	-90	-	dBm
	12 Mbps OFDM	-	-89	-	dBm
	18 Mbps OFDM	-	-87	-	dBm
	24 Mbps OFDM	-	-85	-	dBm
	36 Mbps OFDM	-	-81	-	dBm
	48 Mbps OFDM	-	-77	-	dBm
	54 Mbps OFDM	-	-76	-	dBm
Sensitivity - IEEE 802.11n, 20 MHz (10% PER)	HT20, MCS0	-	-90	-	dBm
	HT20, MCS1	-	-89	-	dBm
	HT20, MCS2	-	-86	-	dBm
	HT20, MCS3	-	-83	-	dBm
	HT20, MCS4	-	-80	-	dBm
	HT20, MCS5	-	-76	-	dBm
	HT20, MCS6	-	-74	-	dBm
	HT20, MCS7	-	-73	-	dBm
Sensitivity - IEEE 802.11n, 40 MHz (10% PER)	HT40, MCS0	-	-88	-	dBm
	HT40, MCS1	-	-86	-	dBm
	HT40, MCS2	-	-83	-	dBm
	HT40, MCS3	-	-81	-	dBm
	HT40, MCS4	-	-77	-	dBm
	HT40, MCS5	-	-73	-	dBm
	HT40, MCS6	-	-72	-	dBm
	HT40, MCS7	-	-70	-	dBm
Sensitivity - IEEE 802.11ac, 20 MHz (10% PER)	VHT20, MCS0	-	-90	-	dBm
	VHT20, MCS1	-	-88	-	dBm
	VHT20, MCS2	-	-85	-	dBm

Parameter	Condition	Min.	Typ.	Max.	Unit
	VHT20, MCS3	-	-82	-	dBm
	VHT20, MCS4	-	-79	-	dBm
	VHT20, MCS5	-	-74	-	dBm
	VHT20, MCS6	-	-73	-	dBm
	VHT20, MCS7	-	-72	-	dBm
Sensitivity - IEEE 802.11ax, 20 MHz (10% PER)	HE20, MCS0	-	-90	-	dBm
	HE20, MCS1	-	-88	-	dBm
	HE20, MCS2	-	-86	-	dBm
	HE20, MCS3	-	-84	-	dBm
	HE20, MCS4	-	-80	-	dBm
	HE20, MCS5	-	-76	-	dBm
	HE20, MCS6	-	-74	-	dBm
	HE20, MCS7	-	-72	-	dBm
	HE20, MCS8	-	-68	-	dBm
	HE20, MCS9	-	-66	-	dBm
Maximum Receive Level					
Maximum receive level @ 5 GHz	11a: 6 Mbps (10% PER, 1000 octets)	-	10	-	dBm
	11a: 54 Mbps (10% PER, 1000 octets)	-	5	-	dBm
	11n: HT20 MCS0 (10% PER, 4096 octets)	-	10	-	dBm
	11n: HT20 MCS7 (10% PER, 4096 octets)	-	0	-	dBm
	11n: HT40 MCS0 (10% PER, 4096 octets)	-	10	-	dBm
	11n: HT40 MCS7 (10% PER, 4096 octets)	-	0	-	dBm
	11ac: VHT20 MCS0 (10% PER, 4096 octets)	-	10	-	dBm

Parameter	Condition		Min.	Typ.	Max.	Unit
	11ac: VHT20 MCS7 (10% PER, 4096 octets)		-	0	-	dBm
	11ax: HE20 MCS0 (10% PER, 4096 octets)		-	10	-	dBm
	11ax: HE20 MCS9 (10% PER, 4096 octets)		-	0	-	dBm
Adjacent Channel Rejection						
Adjacent channel (±20 MHz) rejection - IEEE 802.11a (10% PER with desired signal level as specified in Condition)	6 Mbps OFDM	-79 dBm	16	39	-	dB
	54 Mbps OFDM	-62 dBm	-1	26	-	dB
Adjacent channel (±20 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Condition)	HT20, MCS0	-79 dBm	16	38	-	dB
	HT20, MCS7	-61 dBm	-2	21	-	dB
Adjacent channel (±40 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Condition)	HT40, MCS0	-76 dBm	16	33	-	dB
	HT40, MCS7	-58 dBm	-2	19	-	dB
Adjacent channel (±20 MHz) rejection - IEEE 802.11ac (10% PER with desired signal level as specified in Condition)	VHT20, MCS0	-79 dBm	16	38	-	dB
	VHT20, MCS7	-61 dBm	-2	21	-	dB
Adjacent channel (±20 MHz) rejection - IEEE 802.11ax (10% PER with desired signal level as specified in Condition)	HE20, MCS0	-79 dBm	16	33	-	dB
	HE20, MCS9	-54 dBm	-9	2	-	dB
Nonadjacent Channel Rejection						
Nonadjacent channel (±40 MHz) rejection - IEEE 802.11a (10% PER with desired signal level as specified in Condition)	6 Mbps OFDM	-79 dBm	32	41	-	dB
	54 Mbps OFDM	-62 dBm	15	23	-	dB

Parameter	Condition		Min.	Typ.	Max.	Unit
Nonadjacent channel (± 40 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Condition)	HT20, MCS0	-79 dBm	32	39	-	dB
	HT20, MCS7	-61 dBm	14	19	-	dB
Nonadjacent channel (± 80 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Condition)	HT40, MCS0	-76 dBm	32	38	-	dB
	HT40, MCS7	-58 dBm	14	20	-	dB
Nonadjacent channel (± 40 MHz) rejection - IEEE 802.11ac (10% PER with desired signal level as specified in Condition)	VHT20, MCS0	-79 dBm	32	38	-	dB
	VHT20, MCS7	-61 dBm	14	21	-	dB
Nonadjacent channel (± 40 MHz) rejection - IEEE 802.11ax (10% PER with desired signal level as specified in Condition)	HE20, MCS0	-79 dBm	32	37	-	dB
	HE20, MCS9	-54 dBm	7	11	-	dB

5.5.1.4 WLAN 5 GHz Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	4900	-	5925	MHz
TX Power					
TX power - IEEE 802.11a (EVM compliant)	6 Mbps OFDM	-	17	-	dBm
	54 Mbps OFDM	-	15	-	dBm
TX power - IEEE 802.11n (EVM compliant)	HT20, MCS0	-	16	-	dBm
	HT20, MCS7	-	14	-	dBm
	HT40, MCS0	-	14	-	dBm
	HT40, MCS7	-	12	-	dBm
TX power - IEEE 802.11ac	VHT20, MCS0	-	16	-	dBm

Parameter	Condition	Min.	Typ.	Max.	Unit
(EVM compliant)	VHT20, MCS7	-	14	-	dBm
TX power - IEEE 802.11ax	HE20, MCS0	-	16	-	dBm
(EVM compliant)	HE20, MCS9	-	12	-	dBm

5.5.2 Bluetooth LE RF Characteristics

5.5.2.1 Bluetooth LE Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2402	-	2480	MHz
Bluetooth LE 1 Mbps					
Sensitivity	30.8% PER	-	-97	-	dBm
Maximum input level	30.8% PER	-	10	-	dBm
C/I co-channel	-	-	8	21	dB
C/I 1 MHz adjacent channel	-	-	1	15	dB
C/I -1 MHz adjacent channel	-	-	1	15	dB
C/I 2 MHz adjacent channel	Image frequency	-	-29	-9	dB
C/I -2 MHz adjacent channel	-	-	-30	-17	dB
C/I 3 MHz adjacent channel	Image frequency + 1 MHz	-	-30	-15	dB
C/I -3 MHz adjacent channel	-	-	-41	-27	dB
C/I > 3 MHz adjacent channel	-	-	-38	-27	dB
C/I < -3 MHz adjacent channel	-	-	-52	-27	dB
Out-of-band blocking	30–2000 MHz	-30	-	-	dBm
	2003–2399 MHz	-35	-	-	dBm
	2484–2997 MHz	-35	-	-	dBm
	3000 MHz–12.75 GHz	-30	-	-	dBm
Intermodulation	-	-	-32	-	dBm

Parameter	Condition	Min.	Typ.	Max.	Unit
Bluetooth LE 2 Mbps					
Sensitivity	30.8% PER	-	-94	-	dBm
Maximum input level	30.8% PER	-	10	-	dBm
C/I co-channel	-	-	8	21	dB
C/I 2 MHz adjacent channel	-	-	1	15	dB
C/I -2 MHz adjacent channel	-	-	-2	15	dB
C/I 4 MHz adjacent channel	Image frequency	-	-28	-9	dB
C/I -4 MHz adjacent channel	-	-	-34	-17	dB
C/I 6 MHz adjacent channel	Image frequency + 2 MHz	-	-33	-15	dB
C/I -6 MHz adjacent channel	-	-	-49	-27	dB
C/I > 6 MHz adjacent channel	-	-	-41	-27	dB
C/I < -6 MHz adjacent channel	-	-	-48	-27	dB
Out-of-band blocking	30–2000 MHz	-30	-	-	dBm
	2003–2399 MHz	-35	-	-	dBm
	2484–2997 MHz	-35	-	-	dBm
	3000 MHz–12.75 GHz	-30	-	-	dBm
Intermodulation	-	-	-31	-	dBm
Bluetooth LE 125 kbps					
Sensitivity	30.8% PER	-	-104	-	dBm
Maximum input level	30.8% PER	-	10	-	dBm
C/I co-channel	-	-	3	12	dB
C/I 1 MHz adjacent channel	-	-	-18	6	dB
C/I -1 MHz adjacent channel	-	-	-16	6	dB
C/I 2 MHz adjacent channel	Image frequency	-	-35	-18	dB
C/I -2 MHz adjacent channel	-	-	-42	-26	dB
C/I 3 MHz adjacent channel	Image frequency + 1 MHz	-	-42	-24	dB
C/I -3 MHz adjacent channel	-	-	-46	-36	dB
C/I > 3 MHz adjacent channel	-	-	-47	-36	dB

Parameter	Condition	Min.	Typ.	Max.	Unit
C/I < -3 MHz adjacent channel	-	-	-49	-36	dB
Bluetooth LE 500 kbps					
Sensitivity	30.8% PER	-	-100	-	dBm
Maximum input level	30.8% PER	-	10	-	dBm
C/I co-channel	-	-	4	17	dB
C/I 1 MHz adjacent channel	-	-	-2	11	dB
C/I -1 MHz adjacent channel	-	-	-2	11	dB
C/I 2 MHz adjacent channel	Image frequency	-	-31	-13	dB
C/I -2 MHz adjacent channel	-	-	-31	-21	dB
C/I 3 MHz adjacent channel	Image frequency + 1 MHz	-	-33	-19	dB
C/I -3 MHz adjacent channel	-	-	-42	-31	dB
C/I > 3 MHz adjacent channel	-	-	-39	-31	dB
C/I < -3 MHz adjacent channel	-	-	-54	-31	dB

5.5.2.2 Bluetooth LE Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2402	-	2480	MHz
TX power	-	-20	6	25	dBm
Bluetooth LE 1 Mbps					
In-band emissions	±2 MHz offset	-	-	-50	dBm
	±3 MHz offset	-	-	-51	dBm
	>±3 MHz offset	-	-	-52	dBm
Modulation characteristics	Δf1avg	-	225	245	kHz
	Δf2max	-	185	230	kHz
	Δf2avg/Δf1avg	-	0.8	0.9	-
	Max f _n n = 0, 1, 2, 3...k	-	-	3	150 kHz

Parameter		Condition	Min.	Typ.	Max.	Unit
Carrier frequency offset and drift	Max $ f_0 - f_n $ $n = 2, 3, 4 \dots k$	-	-	2.5	50	kHz
	$ f_1 - f_0 $	-	-	2	23	kHz
	Max $ f_n - f_{n-5} $ $n = 6, 7, 8 \dots k$	-	-	2.5	20	kHz/50 μ s
Bluetooth LE 2 Mbps						
In-band emissions	± 4 MHz offset	-	-	-53	-	dBm
	± 5 MHz offset	-	-	-53	-	dBm
	$> \pm 5$ MHz offset	-	-	-53	-	dBm
Modulation characteristics	Δf_{1avg}	-	450	490	550	kHz
	Δf_{2max}	-	370	470	-	kHz
	$\Delta f_{2avg}/\Delta f_{1avg}$	-	0.8	0.9	-	-
Carrier frequency offset and drift	Max $ f_n $ $n = 0, 1, 2, 3 \dots k$	-	-	3	150	kHz
	Max $ f_0 - f_n $ $n = 2, 3, 4 \dots k$	-	-	2.5	50	kHz
	$ f_1 - f_0 $	-	-	1.5	23	kHz
	Max $ f_n - f_{n-5} $ $n = 6, 7, 8 \dots k$	-	-	2.5	20	kHz/50 μ s
Bluetooth LE 125 kbps						
In-band emissions	± 2 MHz offset	-	-	-49	-	dBm
	± 3 MHz offset	-	-	-51	-	dBm
	$> \pm 3$ MHz offset	-	-	-52	-	dBm
Modulation characteristics	Δf_{1avg}	-	225	245	275	kHz
	Δf_{1max}	-	185	250	-	kHz
Carrier frequency offset and drift	Max $ f_n $ $n = 0, 1, 2, 3 \dots k$	-	-	3	150	kHz
	Max $ f_0 - f_n $ $n = 1, 2, 3 \dots k$	-	-	2.5	50	kHz
	$ f_0 - f_3 $	-	-	1.5	19.2	kHz
	$ f_n - f_{n-3} $ $n = 7, 8, 9 \dots k$	-	-	3	19.2	kHz/48 μ s
Bluetooth LE 500 kbps						
	± 2 MHz offset	-	-	-49	-	dBm

Parameter		Condition	Min.	Typ.	Max.	Unit
In-band emissions	±3 MHz offset	-	-	-51	-	dBm
	>±3 MHz offset	-	-	-52	-	dBm

5.5.3 IEEE 802.15.4 RF Characteristics

5.5.3.1 IEEE 802.15.4 Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
O-QPSK 250 kbps					
Sensitivity	1 % PER	-	-103	-	dBm
Maximum input level	1 % PER	-	10	-	dBm
Co-channel rejection	1 % PER, desired signal @ -82 dBm	-	0	-	dB
Adjacent channel (+5 MHz) rejection	1 % PER, desired signal @ -82 dBm	0	27	-	dB
Adjacent channel (-5 MHz) rejection	1 % PER, desired signal @ -82 dBm	0	35	-	dB
Alternate channel (+10 MHz) rejection	1 % PER, desired signal @ -82 dBm	30	42	-	dB
Alternate channel (-10 MHz) rejection	1 % PER, desired signal @ -82 dBm	30	40	-	dB
Channel (+15 MHz) rejection	1 % PER, desired signal @ -82 dBm	30	44	-	dB
Channel (-15 MHz) rejection	1 % PER, desired signal @ -82 dBm	30	50	-	dB

5.5.3.2 IEEE 802.15.4 Transmitter Characteristics

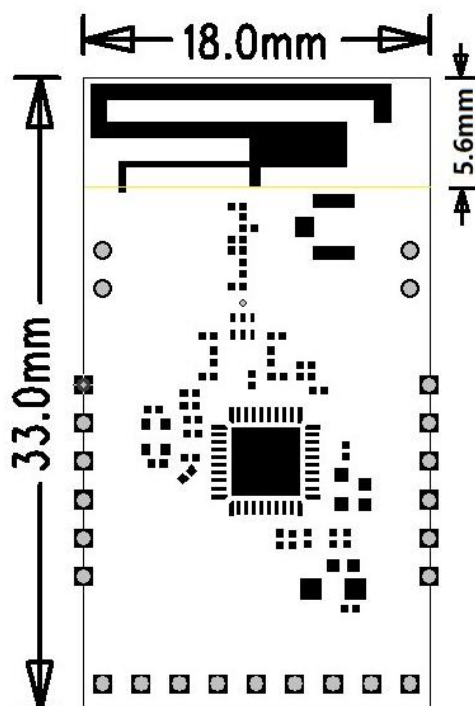
Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Condition	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
O-QPSK 250 kbps					
Maximum output power	-	-20	10	20	dBm
Error-vector magnitude (EVM)	-	-	6	-	%

6 Mechanical Information

6.1 Module Dimensions

Figure 6-1 M39N Dimensions



The M39N dimensions are as below:

- Length: 33.0 mm
- Width: 18.0 mm
- Board thickness: 1.0 mm
- Maximum overall thickness: 2.5 mm

7 Recommended Land Pattern

This section provides the recommended PCB land pattern for surface-mount (SMD) applications and the hole layout for through-hole mounting.

Note: The cross-hatched area in Figure 7-1 indicates the module's exposed pads. Keep-out zones are required in the corresponding areas on the host PCB.

Figure 7-1 Recommended Land Pattern (SMD Mounting) (Unit: mm)

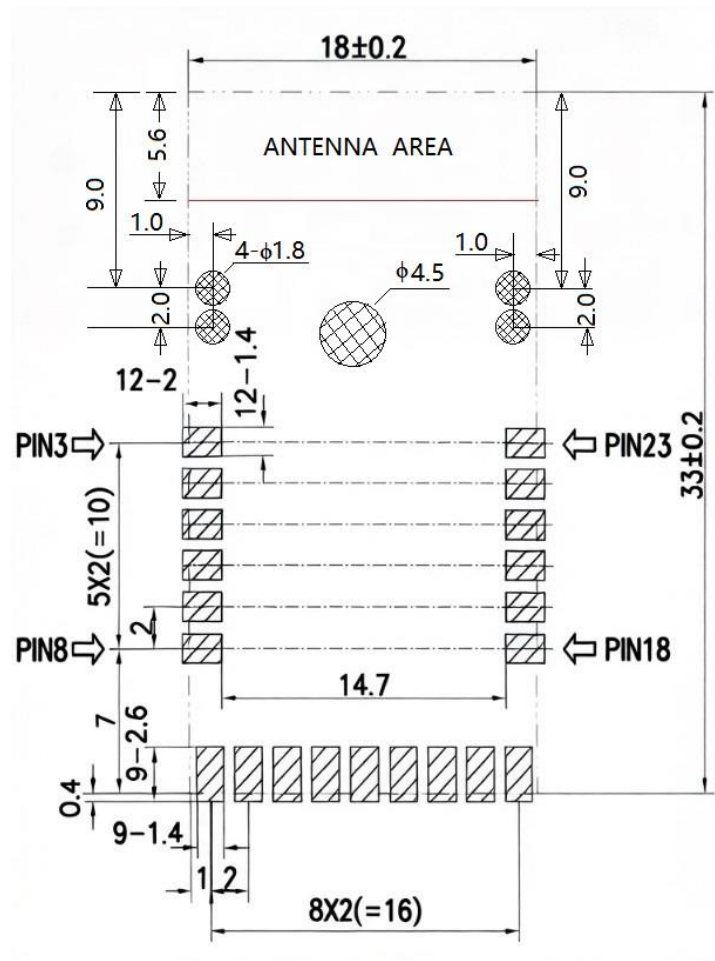
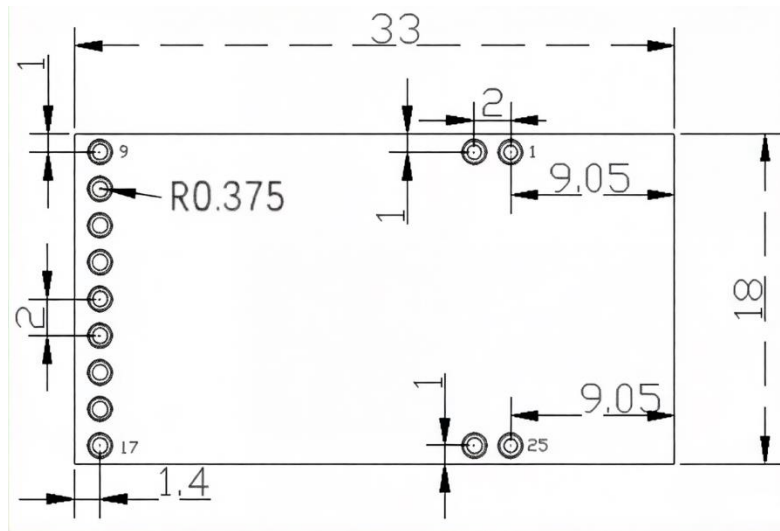


Figure 7-2 Recommended PCB Hole Layout (Through-hole Mounting) (Unit: mm)



8 Product Handling

8.1 Moisture Sensitivity Level (MSL)

The product is applicable to MSL3 in accordance with IPC/JEDEC J-STD-020.

Floor life: After the moisture barrier bag (MBB) is opened, the device shall be stored at $< 30\text{ }^{\circ}\text{C}$ / $< 60\%$ relative humidity (RH) and the product shall be used within 168 hours.

8.2 Storage Conditions

The product shall be stored in a sealed MBB. To ensure product reliability, the following storage conditions must be maintained:

- Pre-opening (in sealed MBB):
 - Storage environment: $< 40\text{ }^{\circ}\text{C}$ and $< 90\%$ RH
 - Calculated shelf life in sealed MBB: 12 months from the seal date
- Post-opening (after unsealing):
 - Refer to the 168-hour floor life limit specified in Section 8.1.

8.3 Baking

The product shall be baked before soldering, if:

- Humidity Indicator Card (HIC) is $> 10\%$ when read at $23 \pm 5\text{ }^{\circ}\text{C}$, or
- The floor life requirements specified in Section 8.1 are not met.

If baking is required, the device may be baked for 24–48 hours at $125 \pm 5\text{ }^{\circ}\text{C}$.

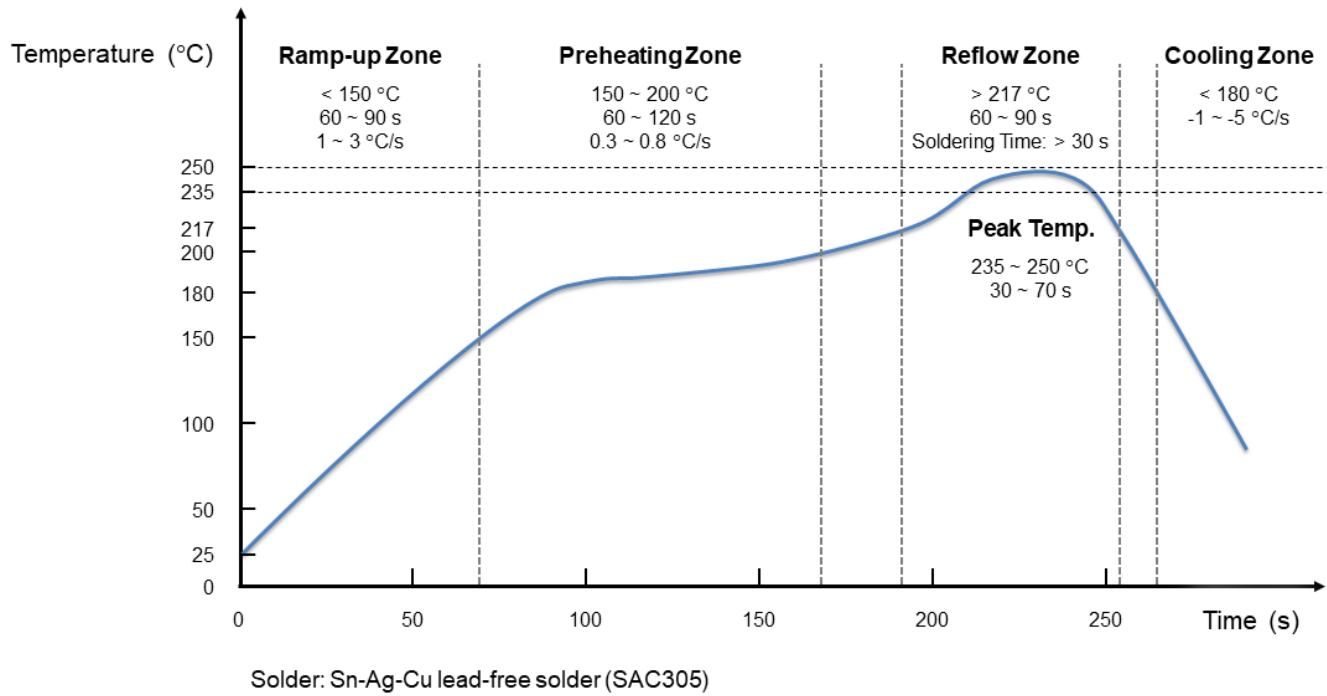
Note: Devices should be placed in high-temperature resistant trays during baking.

8.4 Electrostatic Discharge (ESD)

The product is ESD sensitive and can be damaged by static electricity. Proper ESD techniques should be used when handling the product.

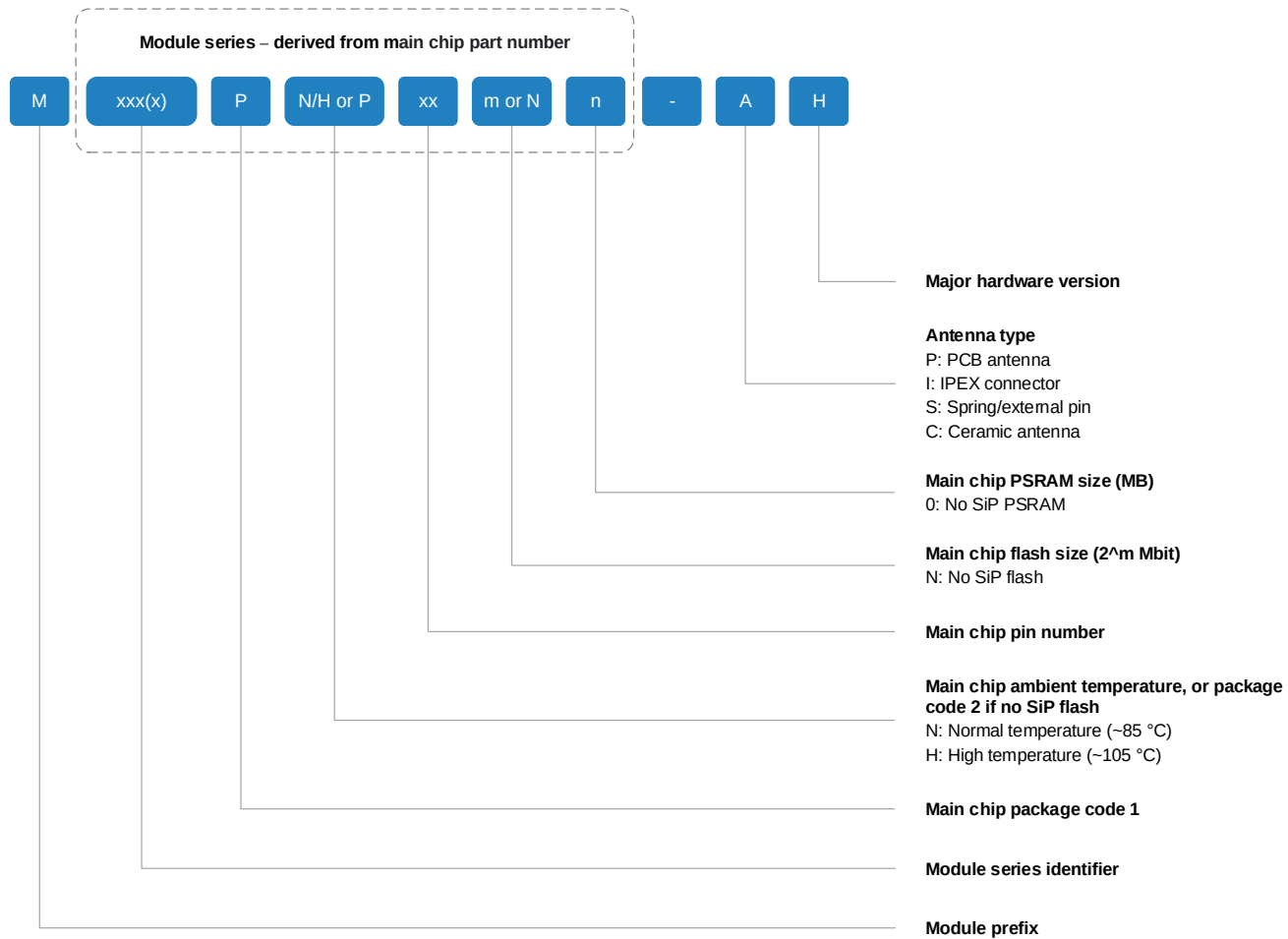
8.5 Reflow Soldering Profile

Figure 8-1 Reflow Soldering Profile



9 Ordering Information

Figure 9-1 Ordering Code Scheme



**Table 9-1 Ordering Information**

Ordering Code	Module Size	Flash	PSRAM	Antenna Type	Embedded Chip	Packing
M39NLQH4050-PA	33.0 mm x 18.0 mm x 2.5 mm	4 MB	-	PCB antenna	BK7239NLQH4050	Tray



Revision History

Version	Date	Description
1.0	2026/3/20	Initial release

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Beken Corporation

Building 41, 1387 Zhangdong Rd
Shanghai 201203
China

<http://www.bekencorp.com>